Elliptic Curve Codes: Recent Constructions and Isogeny-Induced Transformations

Jamie Low Jiaqi & Oliver Ang Wei Jie
School of Physical and Mathematical Sciences, Nanyang Technological University, Singapore

MH4310 Coding Theory, Semester 1, Academic Year 2026–2027

Abstract

Elliptic curves are the only algebraic curves whose points admit a canonical group structure. Algebraic geometry code constructions using elliptic curves can therefore exploit the group law to obtain better parameters and properties. This report introduces the machinery behind elliptic curve codes and surveys recent constructions of MDS elliptic curve codes. As a novel direction, we also explore how isogenies interact with elliptic codes and their parameters, culminating in a new family of quasi-cyclic MDS codes.

1 Introduction

Algebraic geometry (AG) codes are a family of linear error-correcting codes introduced by Valeriy Goppa in the early 1980s [1]. Broadly speaking, they are defined by evaluating rational functions on an algebraic curve $\mathcal{C}/\mathbb{F}_q$ at some set of evaluation points. They generalise the widely used Generalised Reed–Solomon (GRS) codes [2], which are obtained by letting $\mathcal{C}$ be the projective line $\mathbb{P}^1(\mathbb{F}_q)$, which has genus 0.

The code parameters are closely related to the geometry of the chosen curve, particularly its genus and the number of rational points. AG codes therefore provide a natural connection between algebraic geometry and coding theory. In 1982, Tsfasman, Vlăduţ and Zink took this idea further, using curves with many rational points to construct the first code families whose asymptotic parameters surpass the Gilbert–Varshamov bound [3], demonstrating the ability for algebraic geometry to produce asymptotically good codes and sparking further research of AG codes constructed from a diverse range of algebraic curves.

The simplest AG codes inequivalent to GRS codes arise from elliptic curves, which are smooth projective curves of genus 1. They are the only algebraic curves whose points admit a canonical abelian group structure [4, §X.3].[1] AG codes defined using elliptic curves are called elliptic curve codes, or simply elliptic codes. Using the group structure, algebraic constructions unique to genus 1 have been discovered, yielding codes with better parameters (e.g. maximum distance separable (MDS) codes) and desirable properties like iso-duality, with applications in quantum error correction. The group structure also supplies arguments for proving bounds on code parameters.

Elliptic curves also carry isogenies, which are curve morphisms preserving the group structure. Isogenies are heavily used in post-quantum cryptography [5–7], but their applications to coding theory remain largely unexplored. We embark on an AI-assisted project, using Claude and SageMath to explore the different ways in which isogenies can transform an elliptic curve's function field, and hence the parameters of an elliptic code.

The goal of our report is to explain how the remarkable properties of elliptic curves yield unique and excellent constructions of elliptic codes that stand out from general AG codes. In Sections 2 and 3, we introduce elliptic codes as a concrete and accessible instance of AG codes by developing the necessary background underlying the arithmetic of elliptic curves and elliptic codes. We then survey recent constructions and bounds in Sections 4 and 5. We conclude with a documentation of our exploration

[1] Provided they have an $\mathbb{F}_q$-rational point, elliptic curves are the only algebraic curves isomorphic to their own Jacobians and hence are abelian varieties.

on the interaction between isogenies and elliptic codes in Section 6.

2 Preliminaries

We summarise key definitions and results pertaining to elliptic curves and elliptic codes. For a more detailed description with proofs of results, we refer the reader to the following comprehensive standard references, which our exposition is based on:

- Silverman's book [4] for basic arithmetic geometry, with a focus on elliptic curves;
- Stichtenoth's book [8] for algebraic function fields and algebraic geometry (AG) codes.

Henceforth, let $q = p^m$ denote a prime power.

An **elliptic curve** $\mathcal{E}$ over $\mathbb{F}_q$ is a smooth projective curve of genus 1. Equivalently, $\mathcal{E}$ is the set of points $(x, y) \in \overline{\mathbb{F}}_q^2$ satisfying an equation of the form

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6, \quad a_i \in \mathbb{F}_q \quad (1)$$

together with a point at infinity ∞, provided that no point is singular. A point $P \in \mathcal{E}$ is **$\mathbb{F}_q$-rational** if $P \in \mathbb{F}_q^2 \cup \{\infty\}$. The set of $\mathbb{F}_q$-rational points is denoted by $\mathcal{E}(\mathbb{F}_q)$, and the set $\mathcal{E}(\mathbb{F}_q)\setminus\{\infty\}$ of affine $\mathbb{F}_q$-rational points by $\mathcal{E}(\mathbb{F}_q)^*$. Its cardinality is constrained by the Hasse–Weil bound [4, §V.1]

$$|\#\mathcal{E}(\mathbb{F}_q) - (q+1)| \le \lfloor 2\sqrt{q} \rfloor.$$

The points of $\mathcal{E}$ form an abelian group $(\mathcal{E}, \oplus)$ with identity ∞, where the group addition $\oplus$ is governed by the equations in [4, §III.2] and visualised in Figure 1. As is standard in the literature, $[m]P$ denotes the m-fold sum $P \oplus \cdots \oplus P$ for $m \in \mathbb{N}$, and $[-m]P := \ominus[m]P$.

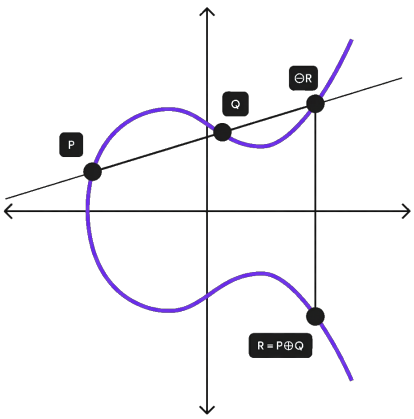


Figure 1. The group law of an elliptic curve $\mathcal{E}$ over $\mathbb{R}$ [9]

A **divisor** of $\mathcal{E}$ is an element D of the free abelian group $\operatorname{Div}\mathcal{E} = \mathbb{Z}[\mathcal{E}]$ over the points of $\mathcal{E}$. In other words, it is a formal sum

$$D = \sum_{P\in\mathcal{E}} n_P \cdot (P), \quad n_P \in \mathbb{Z}$$

where all but finitely many n_P are zero. Its **support** and **degree** are, respectively,

$$\operatorname{supp} D = \{P \in \mathcal{E} : n_P \neq 0\} \quad \text{and} \quad \deg D = \sum_{P\in\mathcal{E}} n_P.$$

Its **sum** is the evaluation of the formal sum using the elliptic group law, and is denoted by

$$\Sigma(D) = \bigoplus_{P\in\mathcal{E}} [n_P]P.$$

A partial ordering on $\operatorname{Div}\mathcal{E}$ is defined by

$$\sum_{P\in\mathcal{E}} n_P(P) \le \sum_{P\in\mathcal{E}} n'_P(P) \iff \forall P \in \mathcal{E} : \quad n_P \le n'_P.$$

The **function field** $\mathbb{F}_q(\mathcal{E})$ of $\mathcal{E}$ is the field $\mathbb{F}_q(x, y)$, where x is transcendental over $\mathbb{F}_q$ and y is algebraic over $\mathbb{F}_q(x)$ subject to Equation (1). For $f \in \mathbb{F}_q(\mathcal{E})$, set

$$\operatorname{div} f := \sum_{P\in\mathcal{E}} \operatorname{ord}_P(f) \cdot (P), \quad \text{where}$$

- $\operatorname{ord}_P(f) = n \in \mathbb{Z}^+$ if f has a zero of order n at P;
- $\operatorname{ord}_P(f) = -n \in \mathbb{Z}^-$ if f has a pole of order n at P;
- $\operatorname{ord}_P(f) = 0$ otherwise.

Such divisors are called **principal** and form a subgroup $\operatorname{PDiv}\mathcal{E}$ of the degree-0 divisors $\operatorname{Div}^0\mathcal{E}$. Two divisors (of any degree) are called **equivalent** if they differ by a principal divisor. The **Jacobian** of $\mathcal{E}$ is the quotient $\operatorname{Jac}\mathcal{E} := \operatorname{Div}^0\mathcal{E}/\operatorname{PDiv}\mathcal{E}$, which is famously isomorphic to $\mathcal{E}$ (cf. Section 4).[2]

Given a divisor $G \in \operatorname{Div}\mathcal{E}$, we construct the **Riemann–Roch space**

$$\mathcal{L}(G) := \{f \in \mathbb{F}_q(\mathcal{E})^* : \operatorname{div} f \ge -G\} \cup \{0\}.$$

This is a finite-dimensional vector subspace of $\mathbb{F}_q(\mathcal{E})$, and we denote its dimension by $\ell(G)$. For elliptic curves, the Riemann–Roch Theorem [8, §1.5] gives us $\ell(G) \ge \deg G$, with equality when $\deg G \ge 1$.

The **space of differential forms** $\Omega_\mathcal{E}$ on $\mathcal{E}$ is the $\mathbb{F}_q$-vector space generated by symbols of the form $\mathrm{d}f$ for $f \in \mathbb{F}_q(\mathcal{E})$, subject to the relations

$$\mathrm{d}(f+g) = \mathrm{d}f + \mathrm{d}g, \quad \mathrm{d}(fg) = f\,\mathrm{d}g + g\,\mathrm{d}f, \quad \mathrm{d}a = 0$$

[2] More precisely, this quotient is called the Picard group of degree 0, but this is isomorphic to the Jacobian variety.

for all $f, g \in \mathbb{F}_q(\mathcal{E})$ and $a \in \mathbb{F}_q$. Around a point $P \in \mathcal{E}$, every $\omega \in \Omega^*_{\mathcal{E}}$ can be written uniquely as

$$\omega = \sum_{n=m}^{\infty} a_n t^n \mathrm{d}t \quad \text{for some} \begin{cases} t \in \mathbb{F}_q(\mathcal{E}),\ \mathrm{ord}_P(t) = 1, \\ a_n \in \mathbb{F}_q,\ a_m \neq 0,\ m \in \mathbb{Z}. \end{cases}$$

For $\omega \in \Omega^*_{\mathcal{E}}$, set $\mathrm{ord}_P(\omega) := m$, $\mathrm{Res}_P(\omega) := a_{-1}$ (or 0 if $m \geq 0$),

$$\mathrm{div}\,\omega := \sum_{P \in \mathcal{E}} \mathrm{ord}_P(\omega) \cdot (P), \quad \text{and}$$
$$\Omega(G) := \{\omega \in \Omega^*_{\mathcal{E}} : \mathrm{div}\,\omega \geq G\} \cup \{0\}.$$

The divisors $\mathrm{div}\,\omega$ are called **canonical**. In genus 1, the sets of canonical and principal divisors coincide [8, Cor. 1.4.12].

3 Elliptic codes

This section introduces general elliptic codes, their parameters, and their encoding and decoding mechanisms. In later sections, we consider specialised codes with better properties.

Let $n \in \mathbb{Z}^+$, $\mathcal{E}$ be an elliptic curve and $D, G \in \mathrm{Div}\,\mathcal{E}$ be two divisors such that $1 \leq \deg G \leq n-1$, $\mathrm{supp}\,D \cap \mathrm{supp}\,G = \varnothing$, and $D = (P_1) + \cdots + (P_n)$ is a sum of n distinct $\mathbb{F}_q$-rational points. We adopt the notation

$$f(D) := (f(P_1), \ldots, f(P_n)) \in \mathbb{F}_q^n \quad \text{for} \quad f \in \mathcal{L}(G).$$

Construct an ordered basis $(\varphi_1, \ldots, \varphi_k)$ of $\mathcal{L}(G)$, where $k = \ell(G) = \deg G$ by Riemann–Roch. Then, there exists an $[n, k, d]_q$ code

$$C_{\mathcal{L}}(D, G) := \{f(D) : f \in \mathcal{L}(G)\} = \langle \varphi_1(D), \ldots, \varphi_k(D) \rangle,$$

which encodes a message $\mathbf{m} \in \mathbb{F}_q^k$ as $\sum_{i=1}^k m_i \varphi_i(D)$.

The quantity $d^* := \deg(D - G) = n - k$ is called the **designed distance** of $C_{\mathcal{L}}(D, G)$. The code has minimum distance $d \geq d^*$ [8, §2], so the Singleton defect is at most 1. We will see in Section 5 ways to construct MDS elliptic codes.

The Euclidean dual code $C_{\mathcal{L}}(D, G)^{\perp_{\mathrm{E}}}$ of $C_{\mathcal{L}}(D, G)$ is

$$C_{\Omega}(D, G) = \left\{(\mathrm{Res}_{P_1}(\omega), \ldots, \mathrm{Res}_{P_n}(\omega)) : \omega \in \Omega(G - D)\right\},$$

as proven in [10, Prop. 8]. Choose $t \in \mathbb{F}_q(\mathcal{E})$ such that $\mathrm{ord}_{P_i}(t) = 1$ for $1 \leq i \leq n$. Then, the differential $\eta := \mathrm{d}t/t$ has the property that every $\omega \in \Omega(G - D)$ can be written as $f\eta$ for some $f \in \mathcal{L}(H)$, where $H = D - G + \mathrm{div}\,\eta$ [8, §8.1]. As such, we have the beautiful result that $C_{\Omega}(D, G) = C_{\mathcal{L}}(D, H)$. Consequently, $C_{\mathcal{L}}(D, G)$ is either MDS or near-MDS.

Common choices of G include $G = k(Q)$ or $G = (k-1)(P) + (Q)$ for some $P, Q \in \mathcal{E}$. Such codes are called **one-point codes** and **two-point codes** respectively. The former has the advantage that a basis of $\mathcal{L}(G)$ is easy to construct.

3.1 Decoding

The remainder of this section is devoted to decoding. We illustrate one relatively simple (but suboptimal) decoding algorithm [11, §6.2][3] for the $[n, n-k, \geq k]$ code $C := C_{\Omega}(D, k(Q))$, the dual of a one-point code.

Denote by $R = \bigcup_{m=1}^{\infty} \mathcal{L}(m(Q))$ the $\mathbb{F}_q$-vector space of functions having no poles away from Q. Let $(\varphi_j)_{j \in \mathbb{Z}^+}$ be a basis for R, ordered such that $\varphi_1 = 1$ and $\mathrm{ord}_Q(\varphi_j) = -j$ for $j \geq 2$, so that $\mathcal{L}(m(Q)) = \langle \varphi_1, \ldots, \varphi_m \rangle$. For instance, if $Q = \infty$, we may take the basis $(1, x, y, x^2, xy, x^3, x^2y, \ldots)$.

Let $\mathbf{r} = \mathbf{c} + \mathbf{e} \in \mathbb{F}_q^n$ be a received word, where $\mathbf{c} \in C$ and $\mathbf{e} \in \mathbb{F}_q^n$ are an unknown codeword and error respectively. Define the **syndrome functional**

$$S : \mathcal{L}(G) \to \mathbb{F}_q, \quad f \mapsto \mathbf{r} \cdot f(D).$$

Let $f, g \in \mathcal{L}(s(Q)) \subset \mathcal{L}(G)$, where $s = \lfloor k/2 \rfloor$, so that $fg \in \mathcal{L}(G)$. Since $(fg)(D) \in C_{\mathcal{L}}(D, G) = C^{\perp}$, we have $\mathbf{c} \cdot (fg)(D) = 0$, and so

$$S(fg) = (\mathbf{c} + \mathbf{e}) \cdot (fg)(D) = \sum_{i \in \mathrm{supp}\,\mathbf{e}} e_i f(P_i) g(P_i).$$

Hence, the space

$$K = \left\{f \in \mathcal{L}(s(Q)) : S(fg) = 0\ \forall\, g \in \mathcal{L}(s(Q))\right\}$$

contains the functions f vanishing at $\mathrm{supp}\,\mathbf{e}$, and thus, reveals the error positions. Sampling $f \in K^*$, the coordinate set I where f vanishes contains $\mathrm{supp}\,\mathbf{e}$. As S is linear, it suffices to consider $S(\varphi_i \varphi_j)$ for $1 \leq i, j \leq s$.

Finally, we use [10, Prop. 6.1] to find the error values. Assuming $\#I < d(C)$, $\mathbf{e}$ is the unique solution $\mathbf{x}$ of the system

$$\mathbf{x} \cdot f_i(D) = S(f_i) \quad \forall\, 1 \leq i \leq k \quad \text{and} \quad x_i = 0 \quad \forall\, i \notin I.$$

[10, Prop. 6.5] shows that this decoding algorithm can correct $\lfloor k/2 \rfloor - 1$ errors (which is suboptimal if C is MDS) with complexity $\mathcal{O}(n^3)$.

[3] This algorithm applies to curves of arbitrary genera and does not use the group law. There are better algorithms [10, 12], but we select this one for pedagogical reasons.

Algorithm 2 (Basic Decoding [11, §6.2])

Input: A received word $\mathbf{r} \in \mathbb{F}_q^n$
Output: The decoded codeword $\mathbf{c} \in C_\Omega(D, G)$
Steps:

1. $\mathbf{S} \leftarrow [S(\varphi_i \varphi_j)]_{i,j=1}^{s} \in \mathbb{F}_q^{s \times s}$
2. $K \leftarrow \ker \mathbf{S}$
3. $f \leftarrow \sum_{i=1}^{s} \lambda_i \varphi_i$ for some $\boldsymbol{\lambda} \in K^*$
4. $I \leftarrow \{i : f(P_i) = 0\}$ // Contains $\operatorname{supp} \mathbf{e}$
5. $\mathbf{H} \leftarrow [\varphi_1(D), \ldots, \varphi_k(D)] \in \mathbb{F}_q^{n \times k}$
6. Let $\mathbf{e}$ be the unique (row) vector satisfying $\mathbf{eH} = \mathbf{rH}$ and $e_i = 0$ for all $i \notin I$
7. Return $\mathbf{r} - \mathbf{e}$

4 Exploiting the group law

Every smooth projective curve $\mathcal{C}$ of genus g carries an abelian variety of dimension g, its Jacobian $\operatorname{Jac}\mathcal{C}$, whose points are the divisor classes of degree zero on $\mathcal{C}$ (cf. Section 2). What distinguishes genus 1 is that the group structure can be transported from the Jacobian to the curve. Specifically, for an elliptic curve, the **Abel–Jacobi map**

$$\mathcal{E} \longrightarrow \operatorname{Jac}\mathcal{E}, \qquad P \longmapsto [(P) - (\infty)]$$

is a bijection whose inverse is the sum map Σ of Section 2, and the group law it induces on $\mathcal{E}$ agrees with the geometric one in [4, §III.3]. This is the isomorphism with the Jacobian noted in the abstract.

In coding theory, one can exploit this group law to obtain better codes. For an AG code $C_\mathcal{L}(D, G)$ with $g = 1$, the points in D and G lie in a group. Conditions on D and G, which govern the code parameters, may therefore be rewritten as conditions on sums of these evaluation points using tools such as the Abel–Jacobi Theorem. For $g \geq 2$ the group lives on a higher-dimensional variety, and no such translation is available.

Lemma 3 (Abel–Jacobi [4, Cor. III.3.5])

$A \in \operatorname{Div}^0 \mathcal{E}$ is principal if and only if $\Sigma(A) = \infty$.

Henceforth we assume, as is standard, that $\operatorname{supp} G \subset \mathcal{E}(\mathbb{F}_q)$, so that $\Sigma(G) \in \mathcal{E}(\mathbb{F}_q)$. Section 3 established that an elliptic code is either MDS or near-MDS. The following criterion decides which.

Theorem 4 (Subset-sum criterion [13, 14])

Let $C_\mathcal{L}(D, G)$ be as in Section 3, with $D = (P_1) + \cdots + (P_n)$ and $k = \deg G$. Then $d = n - k$ if and only if there are k distinct points $P_{i_1}, \ldots, P_{i_k} \in \operatorname{supp} D$ with

$$P_{i_1} \oplus \cdots \oplus P_{i_k} = \Sigma(G). \tag{2}$$

Otherwise, $d = n - k + 1$, and the code is MDS.

The criterion is more than a test, since its proof is constructive and hands us the minimum-weight codewords explicitly. Suppose $S = \{P_{i_1}, \ldots, P_{i_k}\} \subseteq \operatorname{supp} D$ satisfies Equation (2). The divisor $\sum_{P \in S}(P) - G$ then has degree 0 and sum ∞, so Lemma 3 produces a function f with $\operatorname{div} f = \sum_{P \in S}(P) - G$. This f lies in $\mathcal{L}(G)$ and vanishes on S and nowhere else in $\operatorname{supp} D$, so $f(D)$ is a codeword of weight exactly $n - k$. Two functions with the same divisor differ by a constant, and distinct subsets give codewords with distinct supports, so we obtain every minimum-weight codeword exactly once up to scaling. Writing A_{n-k} for the number of codewords of weight $n - k$,

$$A_{n-k} = (q-1) \cdot \#\mathcal{S}, \quad \text{where}$$
$$\mathcal{S} = \left\{ S \subseteq \operatorname{supp} D : \#S = k,\ \bigoplus\nolimits_{P \in S} P = \Sigma(G) \right\}.$$

Counting the codewords of a given weight is hard in general. Here the group law reduces it to counting the solutions of a subset-sum problem in $\mathcal{E}(\mathbb{F}_q)$, which is the form in which the minimum distance of elliptic codes is usually studied [14].

A second consequence concerns the shape of G itself. Replacing G by an equivalent divisor yields an equivalent code. Hence, up to coordinate scalings, we may take G to be supported on up to two points, which is the case for all constructions in this report.

Lemma 5 ([8, Prop. 2.2.14])

If G and H are equivalent divisors whose supports are disjoint from D, then $C_\mathcal{L}(D, G)$ and $C_\mathcal{L}(D, G')$ differ at most by coordinate scaling.

Corollary 6 (Two-point normal form [15, 16])

Fix D and G as in Section 3. Let $Q_0, Q_1 \in \mathcal{E}(\mathbb{F}_q) \setminus \operatorname{supp} D$ be such that $G' := (k-1)(Q_0) +$

(Q_1) satisfies $\Sigma(G') = \Sigma(G)$. Then $C_{\mathcal{L}}(D, G)$ and $C_{\mathcal{L}}(D, G')$ differ at most by coordinate scaling.

A third consequence is that the group law supplies the code with automorphisms. For $R \in \mathcal{E}(\mathbb{F}_q)$, the **translation** map $\tau_R : P \mapsto P \oplus R$ is an automorphism of $\mathcal{E}$ as a curve. It acts on functions by $\tau_R^* f = f \circ \tau_R$ and on divisors by $\tau_R^*(P) = (P \ominus R)$, so that $\tau_R^* \mathcal{L}(G) = \mathcal{L}(\tau_R^* G)$.

Theorem 7 (Translation automorphisms [17])

Let $C := C_{\mathcal{L}}(D, G)$ be as in Section 3. Suppose $R \in \mathcal{E}(\mathbb{F}_q)$ satisfies $[k]R = \infty$ and $\operatorname{supp} D$ is stable under τ_R.[a] Then τ_R induces a monomial automorphism of C.

[a] The latter condition holds, for instance, when $\operatorname{supp} D$ is a subgroup of $\mathcal{E}(\mathbb{F}_q)$ containing R, as in Construction 8.

Elliptic curves have many more curve automorphisms than curves of higher genus [17, 18], and these have been used to construct QC near-MDS codes [19] and optimal locally repairable codes [17].

5 MDS elliptic codes

To determine whether a given AG code is MDS, one usually studies the vanishing of rational functions via Riemann–Roch computations. For $g = 1$, it becomes a purely combinatorial question on a finite abelian group, since $C_{\mathcal{L}}(D, G)$ is MDS precisely when no k-element subset of $\operatorname{supp} D$ sums to $\Sigma(G)$. This section surveys constructions that arrange D and G so that this holds, and then asks how long such a code can be.

Elliptic curves were introduced in order to escape the length restriction on GRS codes, for which $n \leq q + 1$. By the Hasse–Weil bound (cf. Section 2), $\mathcal{E}(\mathbb{F}_q)$ may contain as many as $M := q + 1 + \lfloor 2\sqrt{q} \rfloor$ points. Being MDS, however, is expensive. As $\operatorname{supp} D$ grows, some k-subset of it is eventually forced to sum to $\Sigma(G)$, whence Theorem 4 precludes the MDS property. We shall see below that an MDS elliptic code cannot use more than roughly half of the available rational points. The full length $n \approx q + 2\sqrt{q}$ is available only to near-MDS codes.

5.1 Constructions

The simplest way to obtain the condition of Theorem 4 is a parity argument on a one-point code. Suppose $N := \#\mathcal{E}(\mathbb{F}_q)$ is even and let $H \leq \mathcal{E}(\mathbb{F}_q)$ be a subgroup of index 2, so that $\mathcal{E}(\mathbb{F}_q) = H \sqcup (Q \oplus H)$ for any $Q \notin H$. Sums then behave predictably, as k elements of H sum into H, while k elements of $Q \oplus H$ sum into H or $Q \oplus H$ according to the parity of k.

Construction 8 (Index-two codes [15])

Let $N = \#\mathcal{E}(\mathbb{F}_q)$ be even and let $H \leq \mathcal{E}(\mathbb{F}_q)$ be a subgroup of index 2. Choose $Q \in \mathcal{E}(\mathbb{F}_q) \setminus H$.

(a) If k is odd, set $D = \sum_{P \in H}(P)$ and $G = k(Q)$. Then $C_{\mathcal{L}}(D, G)$ is an $[N/2,\ k,\ N/2 - k + 1]_q$ MDS code.

(b) If k is even, choose $P \in H$ and set $D = \sum_{P' \in H \setminus \{P\}}(P')$ and $G = (k-1)(P) + (Q)$. Then $C_{\mathcal{L}}(D, G)$ is an $[N/2 - 1,\ k,\ N/2 - k]_q$ MDS code.

Proof. In both cases $\operatorname{supp} D \cap \operatorname{supp} G = \varnothing$ and $\deg G = k$, so the stated length and dimension are immediate.

For (a), the quotient $\mathcal{E}(\mathbb{F}_q)/H$ is generated by the coset $Q \oplus H$ and has order 2. As $\Sigma(G) = [k]Q$ with k odd, $\Sigma(G)$ lies in $Q \oplus H$.

For (b), $\Sigma(G) = [k-1]P \oplus Q$ with $[k-1]P \in H$, so again $\Sigma(G)$ lies in $Q \oplus H$.

In both cases, $\Sigma(G) \notin H$. But any k points of $\operatorname{supp} D = H$ sum into H, so Equation (2) fails and $C_{\mathcal{L}}(D, G)$ is MDS by Theorem 4. □

Extending by one coordinate lengthens the code in Construction 8(b) to $[N/2,\ k,\ N/2 - k + 1]_q$ [20], in direct analogy with extended GRS codes. MDS codes of length $N/2$ therefore exist for every admissible k.

A second family of constructions produces MDS codes that are close to self-dual. A code C is **iso-dual** if $C^{\perp_{\mathrm{E}}}$ can be obtained from C only by coordinate scalings, and **self-dual** if $C^{\perp_{\mathrm{E}}} = C$. Both cases force $n = 2k$. The following construction produces iso-dual MDS elliptic codes.[4] Its MDS-ness again follows from Theorem 4, this time by an argument

[4] The original construction [16, Construction 1] was restricted to even characteristic, so that appropriate scalings transform the iso-dual MDS code into self-dual or LCD MDS codes.

about element orders rather than parity.

Construction 9 (Iso-dual MDS codes [16])

Let k and $N = \#\mathcal{E}(\mathbb{F}_q)$ be even, so that $\mathcal{E}(\mathbb{F}_q)$ contains a point Q of order 2. Choose k distinct points $P_1,\dots,P_k \in \mathcal{E}(\mathbb{F}_q)^*$ of odd order where no two are inverse, and set

$$D = \sum_{i=1}^{k}\big((Q\oplus P_i)+(Q\ominus P_i)\big) \quad \text{and}$$
$$G = (k-1)(\infty)+(Q).$$

Then $C_{\mathcal{L}}(D,G)$ is an $[n,k,k+1]_q$ iso-dual MDS code, with $n = 2k \le \frac{N}{2}-1$.

Proof. For the MDS property, let $\mathcal{E}_{\text{odd}} \le \mathcal{E}(\mathbb{F}_q)$ be the subgroup of points of odd order. The sum of any k points $Q\oplus P_{i_j}$ in $\operatorname{supp} D$ is

$$\bigoplus_{j=1}^{k}\big(Q\oplus P_{i_j}\big) = [k]Q\oplus\bigoplus_{j=1}^{k}\big(\oplus P_{i_j}\big) = \bigoplus_{j=1}^{k}\big(\oplus P_{i_j}\big) \in \mathcal{E}_{\text{odd}},$$

since k is even and $[2]Q=\infty$. Since $P_1,\dots,P_k\in\mathcal{E}_{\text{odd}}$, this sum has odd order and cannot equal $\Sigma(G)=Q\notin\mathcal{E}_{\text{odd}}$, so $C_{\mathcal{L}}(D,G)$ is MDS by Theorem 4.

For iso-duality, recall from Section 3 that $C_{\mathcal{L}}(D,G)^{\perp_E} = C_{\mathcal{L}}(D,H)$ with $H = D-G+\operatorname{div}\eta$. Recall from Section 2 that $\operatorname{div}\eta$ is principal. Hence,

$$\deg H = \deg D - \deg G + 0 = 2k-k = k = \deg G.$$

By Lemma 3, $\Sigma(\operatorname{div}\eta)=\infty$. Each pair of points $Q\oplus P_i$ in D sums to $[2]Q=\infty$, so $\Sigma(D)=\infty$ and

$$\Sigma(H) = \Sigma(D)\ominus\Sigma(G)\oplus\Sigma(\operatorname{div}\eta) = \ominus Q = Q = \Sigma(G),$$

the last step using that Q has order 2. By Corollary 6, $C_{\mathcal{L}}(D,H)$ and $C_{\mathcal{L}}(D,G)$ differ at most by coordinate scaling, which gives iso-duality of $C_{\mathcal{L}}(D,G)$.

For the bound on n, $Q\notin\mathcal{E}_{\text{odd}}$ implies $[\mathcal{E}(\mathbb{F}_q):\mathcal{E}_{\text{odd}}]\ge 2$, so $\#\mathcal{E}_{\text{odd}}\le\frac{N}{2}$. Since $\operatorname{supp} D\subseteq Q\oplus(\mathcal{E}_{\text{odd}}\setminus\{\infty\})$, we have $n\le\#\mathcal{E}_{\text{odd}}-1\le\frac{N}{2}-1$. □

5.2 Bounds

Another important question to ask is how long an MDS elliptic code can be. Write $\mathrm{MEC}(k,q)$ for the maximum length of a nontrivial q-ary MDS elliptic code of dimension k, where we temporarily allow $\operatorname{supp} G$ to contain non-rational points. By Theorem 4, the problem is one in additive combinatorics. How large can a subset $\mathcal{T}\subseteq\mathcal{E}(\mathbb{F}_q)$ be before every point of the group is a sum of k distinct elements of $\mathcal{T}$? Once $\operatorname{supp} D$ is that large, no choice of G yields the MDS property.

The first bound is due to Munuera [21], who showed

$$\mathrm{MEC}(k,q) \le \frac{q+1}{2}+\sqrt{q}+k, \tag{3}$$

already confining an MDS elliptic code to about half of the rational points. This was improved by Li, Wan and Zhang [14] for large k, and by Han and Ren [22, 23] for small k, before the exact value was determined by Chen et al. [24], also for small k.

Theorem 10 (Maximal length [24, Table 1])

Let $q\ge 289$ and $3\le k\le\frac{1}{10}\big(q+1-2\sqrt{q}\big)$. Then

$$\mathrm{MEC}(k,q) = \left\lfloor\tfrac{M}{2}\right\rfloor.$$

If we further impose $2\mid k$ and $\operatorname{supp} G\subseteq\mathcal{E}(\mathbb{F}_q)$, the maximal length is instead $\left\lfloor\frac{M}{2}\right\rfloor-1$.

On a maximal curve with N even, so that $N=M$, both constructions of Construction 8 meet this bound, part (b) in its form with $\operatorname{supp} G$ rational.

One may ask whether, given $\mathcal{E}$, we can bound the length of an MDS elliptic code on $\mathcal{E}$ using $\#\mathcal{E}(\mathbb{F}_q)$ instead of the generic Hasse–Weil upper bound M. This is clearly stronger than Theorem 10.

Conjecture 11 (Tighter MDS elliptic bound)

Let C be an MDS $[n,k]_q$ elliptic code on $\mathcal{E}$, with $N=\#\mathcal{E}(\mathbb{F}_q)$. If q (equivalently, N) is sufficiently large and $3\le k\le n-3$, then $n\le\lfloor N/2\rfloor$.[a]

[a] The bound on k excludes trivial cases.

Conjecture 11 is open in general, but proven for iso-dual codes with $N\ge 255$ [16, Thm. 3.1].

6 Isogenies

Let $\mathcal{E}_i$ be an elliptic curve over $\mathbb{F}_q$, for $i=1,2$. An **isogeny** $\varphi:\mathcal{E}_1\to\mathcal{E}_2$ is a nonconstant morphism of curves that fixes ∞. Such a map is automatically a surjective group homomorphism with finite kernel [4, §III.4]. An isogeny induces a **pullback** map φ^* on functions, differentials, and divisors, defined by

$$\begin{aligned}
\mathbb{F}_q(\mathcal{E}_2) &\to \mathbb{F}_q(\mathcal{E}_1), & f &\mapsto f\circ\varphi \\
\Omega_{\mathcal{E}_2} &\to \Omega_{\mathcal{E}_1}, & \mathrm{d}f &\mapsto \mathrm{d}(f\circ\varphi) \\
\operatorname{Div}\mathcal{E}_2 &\to \operatorname{Div}\mathcal{E}_1, & \sum_{Q\in\mathcal{E}_2} n_Q(Q) &\mapsto \sum_{Q\in\mathcal{E}_2} n_Q \sum_{P\in\varphi^{-1}(Q)} e_P(P)
\end{aligned}$$

where e_P denotes the ramification index of $P \in \mathcal{E}_1$ in $\varphi^{-1}(Q)$, the **fibre** over $Q \in \mathcal{E}_2$. Thus, φ induces a field extension $\varphi^*\mathbb{F}_q(\mathcal{E}_2) \le \mathbb{F}_q(\mathcal{E}_1)$, whose degree is denoted by $\ell := \deg\varphi$. An isogeny is called **separable** if this extension is separable, which we assume henceforth.[5] Additionally, there is a **pushforward** map φ_* on divisors and functions, defined by

$$\begin{aligned}
\mathbb{F}_q(\mathcal{E}_1) &\to \mathbb{F}_q(\mathcal{E}_2), & f &\mapsto \left(\varphi_* f : Q \mapsto \prod_{P\in\varphi^{-1}(Q)} f(P)\right) \\
\operatorname{Div}\mathcal{E}_1 &\to \operatorname{Div}\mathcal{E}_2, & \sum_{P\in\mathcal{E}_1} n_P(P) &\mapsto \sum_{P\in\mathcal{E}_1} n_P\big(\varphi(P)\big),
\end{aligned}$$

where $\varphi_* f$ is simply the algebraic norm and $(\varphi_* f)(Q)$ is defined whenever f is regular on $\varphi^{-1}(Q)$. Every isogeny φ comes with a **dual isogeny** $\hat{\varphi} : \mathcal{E}_2 \to \mathcal{E}_1$, which satisfies $\hat{\varphi}\circ\varphi = \varphi_*\circ\varphi^* = [\ell]$ [4, §III.6].

6.1 Exploration

There are several ways that an isogeny φ can transform the code $C_{\mathcal{L}}(D,G)$ into another, namely

1. $C_{\mathcal{L}}(\varphi^*D, \varphi^*G) = \{f(\varphi^*D) : f \in \mathcal{L}(\varphi^*G)\}$,
2. $C_{\mathcal{L}}(\varphi_*D, \varphi_*G) = \{f(\varphi_*D) : f \in \mathcal{L}(\varphi_*G)\}$,
3. $C_{\varphi^*\mathcal{L}}(\varphi^*D, G) = \{(\varphi^*f)(\varphi^*D) : f \in \mathcal{L}(G)\}$, or
4. $C_{\varphi_*\mathcal{L}}(\varphi_*D, G) = \{(\varphi_*f)(\varphi_*D) : f \in \mathcal{L}(G)\}$,

where D and G live on $\mathcal{E}_2$ for the pullbacks and on $\mathcal{E}_1$ for the pushforwards.

This interaction is largely unexplored in the literature. We tasked Claude Opus 5.5 with writing SageMath code that constructs all four codes, and tested it on curves over small fields, notably a 3-isogeny from $\mathcal{E}_1 : y^2 = x^3 + 5$ to $\mathcal{E}_2 : y^2 = x^3 + 20$ over $\mathbb{F}_{31}$, whose 39 rational points form a cyclic group. The code can be found in [25]. Running the experiments, we observe the following.

- Method 4 gives nonlinear codes, as φ_* is multiplicative rather than additive on functions. The experiment with $k = 3$ yielded only 3110 distinct nonzero vectors, whereas a linear $[12,3]_{31}$ code has $31^3 - 1 = 29790$.
- Method 2 gives invalid AG codes if $\operatorname{supp} D$ meets some fibre $\varphi^{-1}(Q)$ more than once, since φ_*D then has a repeated point Q.
- Method 3 gives a subcode of method 1, as $\varphi^*\mathcal{L}(G) \subseteq \mathcal{L}(\varphi^*G)$. Since $(\varphi^*f)(\varphi^*(P)) = f(\varphi_*(\varphi^*P)) = f(\ell(P))$, $C_{\varphi^*\mathcal{L}}(\varphi^*D, G)$ is the ℓ-fold repetition of $C_{\mathcal{L}}(D,G)$, in which each coordinate is repeated along its fibre. The code has parameters $[\ell n, k, \ell d]$. For $k \ge 3$, it is not an elliptic code itself, as $\ell d \le \ell(n-k+1) < \ell n - k$ violates the bound of Section 3.
- Method 1 is the most interesting, and we study it for the rest of the section. It deserves its own definition and a more compact notation.

Definition 12 (Pullback codes)

Let $\varphi : \mathcal{E}_1 \to \mathcal{E}_2$ be a separable isogeny of degree ℓ with cyclic kernel $K := \ker\varphi \subseteq \mathcal{E}_1(\mathbb{F}_q)$. Let $C := C_{\mathcal{L}}(D,G)$ be an $[n,k]_q$ elliptic code on $\mathcal{E}_2$ with

$$D = \textstyle\sum_{i=1}^n (P_i) \in \operatorname{Div}\mathcal{E}_2, \quad \varphi^{-1}(P_i) \subseteq \mathcal{E}_1(\mathbb{F}_q).$$

We call $\varphi^*C := C_{\mathcal{L}}(\varphi^*D, \varphi^*G)$ the **pullback code** of C along φ.

All results in the sequel assume the conditions and notation of Definition 12. We first record three facts from arithmetic geometry that subsequent results rely on.

Lemma 13 (Pullback facts)

(a) $\tau_T^*\varphi^* = \varphi^*$ for every $T \in K$.

(b) $\mathbb{F}_q(\mathcal{E}_1)/\varphi^*\mathbb{F}_q(\mathcal{E}_2)$ is Galois with group $\{\tau_T^* : T \in K\}$, so the functions fixed by every τ_T^* are exactly the pullbacks.

(c) For $A, B \in \operatorname{Div}\mathcal{E}_2$, we have $\varphi^*A \ge \varphi^*B$ if and only if $A \ge B$. A divisor on $\mathcal{E}_1$ whose coefficients are constant on cosets of K is a pullback.

Proof. (a) $\varphi(P \oplus T) = \varphi(P)$ for $T \in K$. (b) Follows from [4, Thm. III.4.10]. (c) As φ is separable, $\varphi^*(Q)$ is the sum of the ℓ distinct points of the fibre over Q. Distinct points have disjoint fibres. Comparing

[5] Equivalently, $\deg\varphi = \#\ker\varphi$, or $e_P = 1\ \forall P$.

coefficients gives the claim. □

6.2 Basic properties

In this subsection, we translate our observations of the code from method 1 into theorems with proofs. We begin with its most obvious properties, namely its parameters and quasi-cyclicity. A code is **quasi-cyclic** of index m (or co-index ℓ) if it is invariant under a coordinate permutation consisting of m disjoint cycles of length ℓ [26].

Theorem 14 (Quasi-cyclic pullback codes)

φ^*C is an $[\ell n, \ell k, \geq \ell(n-k)]_q$ elliptic code on $\mathcal{E}_1$. Furthermore, φ^*C is quasi-cyclic of co-index ℓ.

Proof. The fibres over the points of D form n disjoint cosets of K in $\mathcal{E}_1(\mathbb{F}_q)$, each with size ℓ. Thus, φ^*D is a sum of ℓn distinct rational points, disjoint from $\operatorname{supp}\varphi^*G$. As $\deg\varphi^*G = \ell k$, the code φ^*C is elliptic with the stated parameters by Section 3.

Now, we show that φ^*C is quasi-cyclic of co-index ℓ. Let $\mathbf{c} = f(\varphi^*D) \in \varphi^*C$ be a codeword, with $f \in \mathcal{L}(\varphi^*G)$. Let $K = \langle T\rangle$ and consider τ_T^*. Since

$$(\tau_T^* f)(Q) = f(Q \oplus T) \quad \text{and} \quad Q \oplus T \in \operatorname{supp}\varphi^*D$$

for all $Q \in \operatorname{supp}\varphi^*D$, $(\tau_T^* f)(\varphi^*D)$ is a permutation of $\mathbf{c}$ which cycles points in the same fibre. Since K is cyclic and $\operatorname{ord} T = \ell$, such a permutation clearly consists of n disjoint ℓ-cycles. It remains to show that $\tau_T^* f \in \mathcal{L}(\varphi^*G)$, so that $(\tau_T^* f)(\varphi^*D)$ is a codeword.

By Lemma 13(a), τ_T^* fixes φ^*G. Then,

$$\begin{aligned}\operatorname{div}(\tau_T^* f) + \varphi^*G &= \tau_T^*(\operatorname{div} f) + \tau_T^*(\varphi^*G)\\ &= \tau_T^*(\operatorname{div} f + \varphi^*G) \geq 0,\end{aligned}$$

where the final inequality is due to Lemma 13(c). Hence, $\tau_T^* f \in \mathcal{L}(\varphi^*G)$, and we are done. □

Next, we show that φ^*C inherits iso-duality from C.

Corollary 15 (Iso-dual pullback codes)

If $D - 2G$ is principal, then C and φ^*C are both iso-dual.

Proof. Since $\varphi^*\operatorname{div} f = \operatorname{div}\varphi^* f$, if $D - 2G$ is principal, then so is $\varphi^*D - 2\varphi^*G$. It therefore suffices to show that an elliptic code $C = C_{\mathcal{L}}(D, G)$ with $D - 2G$ principal is iso-dual, as this applies to φ^*C too.

By Section 3, $C^{\perp_E} = C_{\mathcal{L}}(D, H)$ with $H = D - G + \operatorname{div}\eta$. Then, $H - G = D - 2G + \operatorname{div}\eta$ is principal, so $C_{\mathcal{L}}(D, H)$ is a scaling of C by Lemma 5. □

We also observed that the code from method 3 is the subcode of φ^*C fixed by $\ker\varphi$, where elements of $\ker\varphi$ act by translations.

Proposition 16 (Fixed subcode)

The subcode of φ^*C fixed by K is $C_{\varphi^*\mathcal{L}}(\varphi^*D, G)$, which has parameters $[\ell n, k, \ell d]$.

Proof. Let $T \in K$. First, we prove the backward inclusion. By Lemma 13(c), $\varphi^*\mathcal{L}(G) \subseteq \mathcal{L}(\varphi^*G)$, and each $\varphi^* f = f \circ \varphi = f \circ \varphi \circ \tau_T$ is constant on fibres. So, $C_{\varphi^*\mathcal{L}}(\varphi^*D, G) \subseteq \varphi^*C$ is the subcode fixed by K.

Now, we prove the forward inclusion. Suppose $\mathbf{c} := f(\varphi^*D) \in \varphi^*C$ is fixed by K. In other words, the function $\tau_T^* f - f \in \mathcal{L}(\varphi^*G)$ vanishes on φ^*D, which has $\ell n > \deg\varphi^*G$ points. Hence, $\tau_T^* f - f$ must be identically zero, i.e. f is fixed by τ_T^*. By Lemma 13(b), $f \in \varphi^*\mathbb{F}_q(\mathcal{E}_2)$, i.e. $f = \varphi^* g$ for some $g \in \mathbb{F}_q(\mathcal{E}_2)$. But

$$\begin{aligned} f = \varphi^* g \in \mathcal{L}(\varphi^*G) \quad &\Longrightarrow \quad \operatorname{div}\varphi^* g \geq -\varphi^*G\\ \Longrightarrow \quad \operatorname{div} g \geq -G \quad &\Longrightarrow \quad g \in \mathcal{L}(G),\end{aligned}$$

the second implication by Lemma 13(c). Hence, $\mathbf{c} = \varphi^* g(\varphi^*D) \in C_{\varphi^*\mathcal{L}}(\varphi^*D, G)$. □

6.3 Splitting into twists

We asked Claude to construct the generator matrix of φ^*C. Claude did this by examining the eigenspaces of the translation map. The resulting generator matrix was a block matrix consisting of generator matrices for these eigenspaces. We (humans) noticed that this matrix had the structure of the matrix product codes covered in class [20, 27]. We asked Claude to prove this structure in a more general setting. We then verified the proof manually and rewrote it in a clearer fashion. The proof relies on a lemma on the above eigenspaces using tools from arithmetic geometry, which we state first.

Lemma 17 (Eigenfunctions of translations)

In Definition 12, suppose $\ell \mid q-1$ and let $\zeta \in \mathbb{F}_q$ have order ℓ. Write $K = \langle T \rangle$ and $\tau := \tau_T^*$. Then there exist an **eigenfunction** $u \in \mathbb{F}_q(\mathcal{E}_1)^*$ and a divisor $A \in \operatorname{Div}\mathcal{E}_2$ with

(a) $\tau u = \zeta u$, $\operatorname{div} u = \varphi^* A$, $\deg A = 0$ and $\operatorname{supp} A \cap \operatorname{supp} D = \varnothing$;

(b) $R := \Sigma(A)$ generates $\ker\hat{\varphi}$, which is therefore rational;

(c) for every $j \in \mathbb{Z}$, the ζ^j-eigenspace of τ on $\mathcal{L}(\varphi^* G)$ is $u^j \varphi^* \mathcal{L}(G + jA)$.

Due to the length of (and lack of coding theory in) the proof, we postpone it to page 12 in the appendix.

Theorem 18 (Splitting into twists)

Adopt the notation and hypotheses of Lemma 17. Then, $\varphi^* C$ is monomially equivalent to the matrix product code $[C_0, \dots, C_{\ell-1}] \cdot \mathbf{V}$ with $\mathbf{V} := [\zeta^{jm}]_{j,m=0}^{\ell-1}$, where each C_j is a scaling of the twist $C_{\mathcal{L}}(D, G + jA)$.

Up to scaling, one may replace $G + jA$ by any divisor G_j disjoint from D with $\deg G_j = k$ and $\Sigma(G_j) = \Sigma(G) \oplus [j]R$, where R is a suitable generator for $\langle R \rangle = \ker\hat{\varphi}$.

Proof. Write $\tau := \tau_T^*$, and fix $\zeta \in \mathbb{F}_q$ of order ℓ. As τ has order $\ell \nmid q$, the operator τ on $\mathcal{L}(\varphi^* G)$ has minimal polynomial dividing $X^\ell - 1$. This polynomial has roots ζ^j, $0 \le j \le \ell - 1$, so τ is diagonalisable with eigenvalues among the ζ^j. Henceforth, j denotes such an exponent. Let L_j be the ζ^j-eigenspace and W_j its image in $\varphi^* C$. Since evaluation at $\varphi^* D$ is injective, we have $\varphi^* C = \bigoplus_j W_j$.

Let u, A and R be as in Lemma 17, so $L_j = u^j \varphi^* \mathcal{L}(G + jA)$. Fix a choice of $\tilde{P}_i \in \varphi^{-1}(P_i)$ for each i, so that

$$\varphi^{-1}(P_i) = \left\{ \tilde{P}_i \oplus [m]T : m \in \mathbb{Z}/\ell\mathbb{Z} \right\}.$$

Order the coordinates of $\varphi^* C$ (and each W_j) as

$$\tilde{P}_1, \dots, \tilde{P}_n;\ \tilde{P}_1 \oplus T, \dots, \tilde{P}_n \oplus T;\ \tilde{P}_1 \oplus [2]T, \dots, \tilde{P}_n \oplus [2]T;\ \dots;\ \tilde{P}_1 \ominus T, \dots, \tilde{P}_n \ominus T.$$

Any $f \in L_j$ can be expressed as $f = u^j \varphi^* g$ for some $g \in \mathcal{L}(G + jA)$. We have $f(\tilde{P}_i) = u(\tilde{P}_i)^j g(P_i)$, where u has neither a zero nor a pole at $\tilde{P}_i$ as $\operatorname{supp} A \cap \operatorname{supp} D = \varnothing$. Hence, the restrictions of W_j to the coordinates $\tilde{P}_1, \dots, \tilde{P}_n$ form

$$C_j := \mathbf{v}_j \star C_{\mathcal{L}}(D, G + jA) \quad \text{where} \quad \mathbf{v}_j = \left(u(\tilde{P}_i)^j\right)_{i=1}^n.$$

Moreover,

$$f(\tilde{P}_i \oplus [m]T) = (\tau^m f)(\tilde{P}_i) = \zeta^{jm} f(\tilde{P}_i), \quad \text{so}$$
$$W_j = \left\{ \left(\mathbf{c}, \zeta^j \mathbf{c}, \cdots, \zeta^{j(\ell-1)} \mathbf{c}\right) : \mathbf{c} \in C_j \right\}.$$

Hence, $\varphi^* C = \bigoplus_j W_j = [C_0, \dots, C_{\ell-1}] \cdot \mathbf{V}$.

For the second part of the theorem statement, observe that $G + jA$ satisfies the same conditions as G_j, so they are equivalent by Lemma 3. Lemma 5 then asserts that $C_{\mathcal{L}}(D, G_j)$ is a scaling of $C_{\mathcal{L}}(D, G + jA)$. □

Next, we present a necessary condition for $\varphi^* C$ to be MDS. This differs from Theorem 4, which concerns sums of ℓk points as opposed to k points in Corollary 19.

Corollary 19 (MDS twists)

If $\varphi^* C$ is MDS, then no k points of $\operatorname{supp} D$ sum into $\Sigma(G) \oplus \ker\hat{\varphi}$. In the setting of Theorem 18, every twist $C_{\mathcal{L}}(D, G + jA)$ is then MDS.

Proof. By the construction of $\hat{\varphi}$ [4, Thm. III.6.1(b)], every divisor $B \in \operatorname{Div}\mathcal{E}_2$ of degree 0 satisfies $\Sigma(\varphi^* B) = \hat{\varphi}(\Sigma(B))$. Replacing B by $B - b(\infty)$, every divisor B of degree b satisfies

$$\Sigma(\varphi^* B) = \hat{\varphi}(\Sigma(B)) \oplus [b] s_K \quad \text{where} \quad s_K := \Sigma(\varphi^*(\infty)).$$

Suppose that distinct points $P_{i_1}, \dots, P_{i_k} \in \operatorname{supp} D$ satisfy $P_{i_1} \oplus \cdots \oplus P_{i_k} \in \Sigma(G) \oplus \ker\hat{\varphi}$. Applying the identity to both $B = \sum_j (P_{i_j})$ and G, which both have degree k, we obtain

$$\Sigma\left(\varphi^* \textstyle\sum_j (P_{i_j})\right) = \hat{\varphi}\left(P_{i_1} \oplus \cdots \oplus P_{i_k}\right) \oplus [k] s_K$$
$$= \hat{\varphi}(\Sigma(G)) \oplus [k] s_K = \Sigma(\varphi^* G).$$

Since $\varphi^* \sum_j (P_{i_j})$ contains ℓk distinct points of $\operatorname{supp} \varphi^* D$, $\varphi^* C$ is not MDS by Theorem 4. The main result follows by contraposition.

In the setting of Theorem 18, we have $\ker\hat{\varphi} = \langle R \rangle$, so $\Sigma(G + jA) = \Sigma(G) \oplus [j]R$ runs over $\Sigma(G) \oplus \ker\hat{\varphi}$. Hence, if $\varphi^* C$ is MDS, no k points of $\operatorname{supp} D$ sum to any $\Sigma(G + jA)$, and each twist $C_{\mathcal{L}}(D, G + jA)$ is MDS by Theorem 4. □

6.4 Comparison with previous work

We found no publications discussing the application of isogenies to elliptic curve codes. The most similar publication is by Tong and Ding [19], which also constructed quasi-cyclic near-MDS codes, but based on the group automorphisms of $\mathcal{E}$. There are fundamental differences between the two constructions, which we state below in the format: [19] *vs* ours.

- **Symmetry:** group automorphisms of $\mathcal{E}$ *vs* translations by $\ker\varphi$
- **Co-index:** ≤ 6 *vs* any $\ell \mid N$
- G**:** $k(\infty)$ *vs* a pullback divisor
- **Orbit sizes:** sometimes varied *vs* uniform (size ℓ)
- **MDS:** exclusively near-MDS *vs* possibly MDS

6.5 Applications

The results in the previous subsections have three immediate applications. The first is **longer iso-dual elliptic codes**. The hypothesis of Corollary 15 holds for the iso-dual codes of Construction 9 as $\Sigma(D) = \infty = [2]Q = \Sigma(2G)$. Method 1 therefore turns a short iso-dual MDS code of length $2k < \frac{N}{\ell}$ into a long iso-dual code of length $2\ell k$ with the same rate $\frac{1}{2}$ and relative designed distance $\frac{n-k}{n}$, but typically at the cost of the MDS property (i.e. 1 unit of distance).

The second is **decoding via twists**. Ling and Solé showed that every quasi-cyclic code decomposes into shorter constituent codes [26]. Theorem 18 identifies the constituents of φ^*C as scalings of the twists $C_{\mathcal{L}}(D, G_j)$, which are $[n, k, \geq n-k]$ elliptic codes on $\mathcal{E}_2$. Decoding the ℓ twists separately, each up to half its minimum distance, corrects every error pattern meeting at most $\lfloor (n-k-1)/2 \rfloor$ fibres $\varphi^{-1}(P_i)$, however many errors each fibre contains. Thus, errors concentrated on few fibres can be corrected by ℓ decoders of length n, rather than one of length ℓn.

The third is a supply of **quasi-cyclic elliptic codes**. Quasi-cyclic codes generalise cyclic codes and have a rich algebraic structure [26]. By Theorem 14, the pullback produces quasi-cyclic elliptic codes of co-index ℓ with the same rate $\frac{k}{n}$ and relative designed distance $\frac{n-k}{n}$ as C, provided $\mathcal{E}_1(\mathbb{F}_q)$ has a cyclic subgroup of order ℓ. We can combine this with Construction 8 to obtain **MDS quasi-cyclic elliptic codes**, provided that the index-two subgroup is compatible with the fibres of φ. This is illustrated in Construction 20 and Example 21.

Construction 20 (MDS QC pullback codes)

Let $\varphi : \mathcal{E}_1 \to \mathcal{E}_2$ and K be as in Theorem 14, with ℓ odd. Let $N = \#\mathcal{E}_1(\mathbb{F}_q)$ be even, let $H \leq \mathcal{E}_1(\mathbb{F}_q)$ be a subgroup of index 2, and choose $Q \in \mathcal{E}_1(\mathbb{F}_q) \setminus H$.

(a) If k is odd, set $D = \sum_{P \in \varphi(Q \oplus H)}(P)$ and $G = k(\infty)$. Then $\varphi^*C = C_{\mathcal{L}}(\varphi^*D, \varphi^*G)$ is an $[N/2, \ell k, N/2 - \ell k + 1]_q$ MDS code.

(b) For any k, set $D = \sum_{P \in \varphi(H) \setminus \{\infty\}}(P)$ and $G = (k-1)(\infty) + (\varphi(Q))$. Then φ^*C is an $[N/2 - \ell, \ell k, N/2 - \ell - \ell k + 1]_q$ MDS code.

In both cases, φ^*C is quasi-cyclic of co-index ℓ.

Proof. See page 13 in the appendix. □

Example 21 ($[140, 5k]_{251}$ MDS QC codes [25])

Consider $\mathcal{E}_1 : y^2 = x^3 + x + 100$ over $\mathbb{F}_{251}$. Then, $N = \#\mathcal{E}_1(\mathbb{F}_{251}) = 280$, which is 3 off from the upper Hasse–Weil bound $q + 1 + \lfloor 2\sqrt{q} \rfloor = 283$. $(199, 0)$ is its only rational point of order 2, so $H := [2]\mathcal{E}_1(\mathbb{F}_{251})$ has index 2.

The point $T = (65, 14)$ has order $\ell = 5$, and Vélu's formulae give the isogeny $\varphi : \mathcal{E}_1 \to \mathcal{E}_2 : y^2 = x^3 + 231x + 13$ with kernel $K = \langle T \rangle$. As $5 \mid 250 = q - 1$, we may take $\zeta = 20$, an element of order 5 in $\mathbb{F}_{251}$. Then

$$u = \frac{xy + 195x^2 + 99y + 215x + 232}{(x - 65)(x - 204)}$$

is an eigenfunction of $\tau := \tau_T^*$ as in Lemma 17, with $\tau u = \zeta u$. Its poles are the five points of K, and its zeros form the fibre over $R := (141, 209) \in \mathcal{E}_2$, so $\operatorname{div} u = \varphi^*A$ with $A = (R) - (\infty)$.

The 28 points of $\mathcal{E}_2$ whose fibres lie in $\mathcal{E}_1(\mathbb{F}_{251}) \setminus H$ are

(5,±17), (23,±26), (56,±8), (60,±31), (77,±52), (104,±24), (142,±124), (143,±67), (161,±100), (222,±7), (224,±85), (225,±5), (228,±75), and (232,±78).

Let D be their sum (ordered as above, with (x,y) before $(x,-y)$) and $G = k(\infty)$ with k odd. By Construction 20, φ^*C is a quasi-cyclic $[140, 5k, 141-5k]_{251}$ MDS code of co-index 5.

Next, we express φ^*C explicitly as a matrix-product code. Take $\tilde{P}_i$ to be the point of $\varphi^{-1}(P_i)$ with the lexicographically smallest coordinates, and order φ^*D as in the proof of Theorem 18. The twisted divisors are $G + jA = (k-j)(\infty) + j(R)$. Then, φ^*C is the matrix product code $(C_0, \ldots, C_4) \cdot \mathbf{V}$ with

$$\mathbf{V} = \left[\zeta^{jm}\right]_{j,m=0}^{4} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 20 & 149 & 219 & 113 \\ 1 & 149 & 113 & 20 & 219 \\ 1 & 219 & 20 & 113 & 149 \\ 1 & 113 & 219 & 149 & 20 \end{pmatrix} \quad \text{and}$$

$$C_j = \mathbf{v}_j \star C_{\mathcal{L}}(D, G + jA), \quad \mathbf{v}_j := \left(u(\tilde{P}_i)^j\right)_{i=1}^{28}.$$

By Corollary 19, every twist is a $[28, k, 29-k]_{251}$ MDS code. For instance, for $k = 15$, φ^*C is a $[140, 75, 66]_{251}$ code that glues together five $[28, 15, 14]_{251}$ MDS codes.

Taking instead D to be the sum of the 55 affine points of $\varphi(\mathcal{E}_1(\mathbb{F}_{251}))$, Theorem 14 gives quasi-cyclic $[275, 5k]_{251}$ codes for $1 \le k \le 54$. By Equation (3), the codes are near-MDS with minimum distance $275 - 5k$.

7 Conclusion

This report has demonstrated that amongst all AG codes, the genus 1 case possesses unique properties that make it stand out from other genera. The group law of an elliptic curve supplies many beautiful arguments that are used to achieve better code parameters and desirable code properties. We summarise these below.

- Showing whether a code is MDS reduces to the subset-sum criterion (Theorem 4), an algebraic combinatorial problem. This was used to construct MDS codes (Constructions 8, 9 and 20).
- Up to equivalence, we need only consider one- or two-point codes (Corollary 6).
- Code automorphisms are abundant, arising from translations (Theorem 7) as well as from $\mathrm{Aut}(\mathcal{E}, \infty)$.
- Pulling back along an isogeny produces longer elliptic codes that preserve the rate and relative designed distance, inherit iso-duality, are quasi-cyclic, and split into twisted elliptic codes (Section 6).

However, the current research landscape focusing specifically on elliptic codes is relatively limited, compared to other families of AG codes. Open questions include

- efficient and effective decoding,
- the resolution of Conjecture 11, and
- hull variation on the pullback code.

References

[1] V. D. Goppa, "Algebraico-geometric codes," *Mathematics of the USSR-Izvestiya*, vol. 21, no. 1, pp. 75–91, 1983.

[2] F. J. MacWilliams and N. J. A. Sloane, *The theory of error-correcting codes*, ser. North-Holland Mathematical Library. Amsterdam: North-Holland, 1977, vol. 16.

[3] M. A. Tsfasman, S. G. Vlăduţ, and T. Zink, "Modular curves, Shimura curves, and Goppa codes, better than Varshamov–Gilbert bound," *Mathematische Nachrichten*, vol. 109, no. 1, pp. 21–28, 1982.

[4] J. H. Silverman, *The arithmetic of elliptic curves*, 2nd ed., ser. Graduate Texts in Mathematics. Springer, 2009, vol. 106.

[5] J. J. Low, "Genus two isogeny cryptography," In production, Bachelor's Thesis, Nanyang Technological University, Singapore, 2027.

[6] A. Basso, G. Borin, W. Castryck, M. Corte-Real Santos, R. Invernizzi, A. Leroux, L. Maino, F. Vercauteren, and B. Wesolowski, "Prism with a pinch of salt: Simple, efficient, and strongly unforgeable signatures from isogenies," *Journal of Cryptology*, vol. 39, no. 30, 2026.

[7] M. A. Aardal, G. Adj, D. F. Aranha, *et al.*, "SQIsign," National Institute of Standards and Technology, Tech. Rep., 2026. [Online]. Available: https://sqisign.org.

[8] H. Stichtenoth, *Algebraic function fields and codes*, 2nd ed., ser. Graduate Texts in Mathematics. Springer, 2009, vol. 254.

[9] Elusiv Privacy. (2023). An introduction to elliptic curve cryptography, [Online]. Available: https://medium.com/@elusivprivacy/an-introduction-to-elliptic-curve-cryptography-19a6e5752fcf (visited on 10/04/2026).

[10] M. E. O'Sullivan and M. Bras-Amorós, "The key equation for one-point codes," in *Advances in algebraic geometry codes*, World Scientific, 2008, pp. 99–152.

[11] T. Høholdt, J. H. Van Lint, and R. Pellikaan, "Algebraic geometry codes," in *Handbook of coding theory*, V. S. Pless and W. C. Huffman, Eds., vol. 1, Amsterdam: Elsevier, 1998, pp. 871–961.

[12] T. Høholdt and R. Pellikaan, "On the decoding of algebraic-geometric codes," *IEEE Transactions on Information Theory*, vol. 41, no. 6, pp. 1589–1614, 1995.

[13] C. Munuera, "On MDS elliptic codes," *Discrete Mathematics*, vol. 117, no. 1-3, pp. 279–286, 1993.

[14] J. Li, D. Wan, and J. Zhang, "On the minimum distance of elliptic curve codes," in *2015 IEEE International Symposium on Information Theory (ISIT)*, IEEE, 2015, pp. 2391–2395.

[15] P. Wang, W. Liu, J. Luo, and D. Zhai, *Non Reed–Solomon type MDS codes from elliptic curves*, arXiv:2509.04247, 2025.

[16] Y. Zhu and C.-A. Zhao, "On iso-dual MDS codes from elliptic curves," *Finite Fields and Their Applications*, vol. 109, p. 102 699, 2026.

[17] L. Ma and C. Xing, "The group structures of automorphism groups of elliptic curves over finite fields and their applications to optimal locally repairable codes," *Journal of Combinatorial Theory, Series A*, vol. 193, p. 105 686, 2023.

[18] H. Stichtenoth, "Über die Automorphismengruppe eines algebraischen Funktionenkörpers von Primzahlcharakteristik. I. Eine Abschätzung der Ordnung der Automorphismengruppe," *Archiv der Mathematik*, vol. 24, pp. 527–544, 1973.

[19] H. Tong and Y. Ding, "Quasi-cyclic NMDS codes," *Finite Fields and Their Applications*, vol. 24, pp. 45–54, 2013.

[20] M. F. Ezerman and S. Ling, *Lecture Deck 7*, MH4310 Coding Theory, Nanyang Technological University, Singapore, 2026.

[21] C. Munuera, "On the main conjecture on geometric MDS codes," *IEEE Transactions on Information Theory*, vol. 38, no. 5, pp. 1573–1577, 1992.

[22] D. Han and Y. Ren, "A tight upper bound for the maximal length of MDS elliptic codes," *IEEE Transactions on Information Theory*, vol. 69, no. 2, pp. 819–822, 2023.

[23] ——, "The maximal length of q-ary MDS elliptic codes is close to $q/2$," *International Mathematics Research Notices*, vol. 2024, no. 11, pp. 9036–9043, 2024.

[24] H. Chen, C. Hu, J. Huang, and C.-A. Zhao, *On the maximal length of MDS elliptic codes*, arXiv:2605.29439, 2026.

[25] J. J. Low and O. W. J. Ang. (2026). MH4310 elliptic codes, [Online]. Available: https://github.com/jamie-meows04/MH4310-Elliptic-Codes (visited on 10/16/2026).

[26] S. Ling and P. Solé, "On the algebraic structure of quasi-cyclic codes I: Finite fields," *IEEE Transactions on Information Theory*, vol. 47, no. 7, pp. 2751–2760, 2001.

[27] T. Blackmore and G. H. Norton, "Matrix-product codes over $\mathbb{F}_q$," *Applicable Algebra in Engineering, Communication and Computing*, vol. 12, no. 6, pp. 477–500, 2001.

Appendix

Proof of Lemma 17. (a) By Dedekind's lemma, $1, \tau, \ldots, \tau^{\ell-1}$ are linearly independent over $\mathbb{F}_q(\mathcal{E}_1)$, and so over $\varphi^*\mathbb{F}_q(\mathcal{E}_2)$. Thus, τ has minimal polynomial $X^\ell - 1$ on $\mathbb{F}_q(\mathcal{E}_1)$. Hence, there is $u \in \mathbb{F}_q(\mathcal{E}_1)^*$ with $\tau(u) = \zeta u$. Since

$$\tau(\operatorname{div} u) = \operatorname{div}(\tau(u)) = \operatorname{div}(\zeta u) = \operatorname{div} u,$$

$\operatorname{div} u$ is fixed by K. Lemma 13(c) gives $\operatorname{div} u = \varphi^* A$ for some $A \in \operatorname{Div} \mathcal{E}_2$. Also, u^ℓ is fixed by τ, so $u^\ell =$

$\varphi^* h$ for some $h \in \mathbb{F}_q(\mathcal{E}_2)$. Taking div on both sides gives

$$\operatorname{div}(u^\ell) = \ell \operatorname{div} u = \ell \varphi^* A = \varphi^*(\ell A) = \varphi^* \operatorname{div} h,$$

which forces $\ell A = \operatorname{div} h$. In particular, $\deg A = 0$.

By the approximation theorem [8, Thm. 1.3.1], replacing u by $u \cdot \varphi^* a$ for suitable $a \in \mathbb{F}_q(\mathcal{E}_2)^*$, we may assume $\operatorname{supp} A \cap \operatorname{supp} D = \varnothing$.

(b) Put $R := \Sigma(A) \in \mathcal{E}_2(\mathbb{F}_q)$. As $\varphi^* A = \operatorname{div} u$ is principal and $\deg A = 0$, we have $\hat{\varphi}(\Sigma(A)) = \Sigma(\varphi^* A) = \infty$ by Lemma 3, so $R \in \ker \hat{\varphi}$.

We show that R has order ℓ. Suppose $[m]R = \infty$ with $m < \ell$. Then, $mA = \operatorname{div} a$ for some a. Consider the function $v := u^m / \varphi^* a$. Its divisor is $\operatorname{div} v = m\varphi^*(A) - \varphi^*(mA) = 0$, so v is a nonzero constant, hence fixed by τ. But $\tau(v) = (\zeta u)^m / \varphi^* a = \zeta^m v$, which forces $\zeta^m = 1$, a contradiction. Thus, R has order ℓ.

As $\hat{\varphi} \circ \varphi = [\ell]$ is separable for $\ell \nmid q$, so is $\hat{\varphi}$, and $\#\ker \hat{\varphi} = \ell$. Hence R generates $\ker \hat{\varphi}$, which is therefore rational.

(c) Let $f \in \mathcal{L}(\varphi^* G)$, so $\operatorname{div} f \geq -\varphi^* G$. f lies in the ζ^j-eigenspace if and only if f/u^j is fixed by τ, if and only if $f = u^j \varphi^* g$ for some $g \in \mathbb{F}_q(\mathcal{E}_2)$. This already shows that $u^j \varphi^* \mathcal{L}(G + jA)$ lies in the ζ^j-eigenspace. For the reverse inclusion, note that Lemma 13(c) gives $\operatorname{div} f = \varphi^*(jA + \operatorname{div} g) \geq -\varphi^* G$, so $g \in \mathcal{L}(G + jA)$. □

Proof of Construction 20. We check that the codes C and $\varphi^* C$ are well-defined, i.e. the supports of D and G (resp. $\varphi^* D$ and $\varphi^* G$) are disjoint and $\mathbb{F}_q$-rational.

The image of K in $\mathcal{E}_1(\mathbb{F}_q)/H \cong \mathbb{Z}/2\mathbb{Z}$ has order dividing both $\#K = \ell$ and 2, so it is trivial, and $K \subseteq H$. Hence, H and $Q \oplus H$ are unions of cosets of K, i.e. they are partitioned by the fibres of φ. So $\varphi^{-1}(\varphi(H)) = H$, $\varphi^{-1}(\varphi(Q \oplus H)) = Q \oplus H$, and $\varphi(H) \cap \varphi(Q \oplus H) = \varnothing$. In (a), this shows that

$$\operatorname{supp} \varphi^* G = \varphi^{-1}(\infty) = K \subseteq \mathcal{E}_1(\mathbb{F}_q) \quad \text{and}$$
$$\operatorname{supp} \varphi^* D = \varphi^{-1}(\varphi(Q \oplus H)) \subseteq Q \oplus H \subseteq \mathcal{E}_1(\mathbb{F}_q)$$

are $\mathbb{F}_q$-rational. Likewise, in (b), both

$$\operatorname{supp} \varphi^* G = K \cup (Q \oplus K) \subseteq \mathcal{E}_1(\mathbb{F}_q) \quad \text{and}$$
$$\operatorname{supp} \varphi^* D = \varphi^{-1}(\varphi(H \setminus \{\infty\})) \subseteq H \subseteq \mathcal{E}_1(\mathbb{F}_q)$$

are $\mathbb{F}_q$-rational. As $\infty = \varphi(\infty) \in \varphi(H)$ and $\varphi(Q) \in \varphi(Q \oplus H)$, the supports of D and G are disjoint in both cases, so C is well-defined. It follows that the supports of $\varphi^* D$ and $\varphi^* G$ are also disjoint, so $\varphi^* C$ is well-defined.

Now, we show that $\varphi^* C$ is MDS by means of Theorem 4. Write $K = \langle T \rangle$ and $s_K := \sum_{T' \in K} T' = [\ell(\ell - 1)/2]T$. Since ℓ is odd, $s_K = \infty$.

For (a), $\Sigma(\varphi^* G) = [k] s_K = \infty \in H$. But $\operatorname{supp} \varphi^* D = Q \oplus H$, so any ℓk of its points sum into $[\ell k]Q \oplus H = Q \oplus H$, as ℓk is odd.

For (b), the fibre over $\varphi(Q)$ is $Q \oplus K$, so $\Sigma(\varphi^* G) = [k-1]s_K \oplus [\ell]Q \oplus s_K = [\ell]Q$, which lies in $Q \oplus H$ as ℓ is odd. But $\operatorname{supp} \varphi^* D \subseteq H$, so any ℓk of its points sum into H.

In both cases, no ℓk points of $\operatorname{supp} \varphi^* D$ sum to $\Sigma(\varphi^* G)$, so $\varphi^* C$ is MDS by Theorem 4. □

AI prompts

At the beginning, we used Claude to evaluate our potential project directions. We then drafted a section-by-section outline of the report, which Claude referenced to search for relevant and recent publications, suggest results to include, evaluate their suitability for the report, and collate a bibliography containing those results.

- "Sort these project directions on elliptic codes into foundational, doable at undergraduate level, stretch, and novel, with research questions for each."
- "Validate or disprove our comments with research. We want to survey recent constructions such as isodual elliptic codes, study conditions for an elliptic code to be MDS, and include a short exploration of isogenies."
- "Suggest a few constructions that exploit the group law and are easiest to explain, avoiding references that need too much machinery. Motivate them by the fact that only elliptic curves have a group law."

Sections 1 to 3 were fully human-written (other than grammatical fixes). For Sections 4 and 5, we first reviewed the original papers that introduced the constructions that Claude suggested, and selected a few relevant ones. We then wrote a skeleton of

Sections 4 and 5 based on those constructions and used Claude to expand the draft by filling in details such as theorems and references. Of course, we reviewed and revised the drafts.

- "Proofread Sections 1 to 3, and expand the draft of Sections 4 and 5 in the same tone and style."
- "Add one or two paragraphs to Section 4 on code automorphisms arising from the group law."

Ideation for Section 6 was done without AI assistance. After hypothesising the outcome of the four methods, we compared our hypotheses against Claude's own predictions. Claude then generated SageMath code to construct our codes and test our hypotheses. Claude was asked to derive a generator matrix for φ^*C, from which we observed a matrix product pattern and asked it to prove it. Claude also wrote code to compute Example 21.

- "Predict the outcome of each of the four isogeny transformations, and suggest how to analyse them in SageMath, with a sample code file to test each one."
- "Suggest applications of method 1."
- "How does the generator matrix of the code produced by method 1 look like?"

We fed the statement and proof of Lemma 17 and Theorem 18 into ChatGPT and Gemini as a preliminary verification of whether the proof was valid. Thereafter, we (humans) manually verified the proof and refined it to make certain arguments clearer.

We used Claude to proofread our manuscript for errors and missing hypotheses in results.

- "Why does Conjecture 11 require $3 \leq k \leq n-3$, and why does the bound in [16] differ from $\mathrm{MEC}(k,q)$?"
- "Confirm that the rewritten statement of Theorem 18 omits no crucial hypothesis, and scan the literature to determine how novel the proof is."
- "Proofread the changes to the document, including Section 6."

Lastly, we used Claude Code to expedite the process of writing and debugging the Manim code for our presentation.

- "Next slide. Unwrite all text, leaving only the curve. Then, write a Tex "AG Code", bolded, in the top left. Add 3D axes centred at the curve. Rotate the camera to give a 3D view of the graph."
- "Next slide. Animate the points on E1 moving to their images on E2 under the chosen isogeny φ. Add green lines between the image points as before to show that the group law is preserved."